

Part X — Applied Physics · Chapter 54

Physics in Medicine

How physical principles power diagnosis, imaging, treatment, and biomedical engineering

Every heartbeat is a wave of electrical potential that can be measured on the skin. Every X-ray image is a map of how tissue absorbs ionising radiation. Every MRI scan is a response of hydrogen nuclei to a powerful magnetic field. Every ultrasound image is built from the echoes of sound waves reflecting off tissue boundaries. Physics does not merely support medicine as a background technology — it is the foundation on which modern diagnostics and treatment are built. This chapter connects the mechanics, fluid dynamics, thermodynamics, electricity, waves, and nuclear physics you have studied to the instruments and methods that keep people alive and healthy. Understanding medical physics makes you a more informed patient, a more capable future scientist, and a clearer thinker about the technology that defines contemporary healthcare.

54.1 Why Medicine Depends on Physics

Modern medicine is inseparable from physics. Every major diagnostic tool — stethoscope, X-ray, CT scanner, MRI, ultrasound, ECG, pulse oximeter — operates on physical principles. Every treatment technology — radiation therapy, laser surgery, lithotripsy, pacemaker, defibrillator — is designed using physics. The human body itself is a physical system: it obeys Newton's laws, conservation of energy, fluid mechanics, electrostatics, thermodynamics, and wave physics.

Medical physics is the branch of applied physics concerned with the application of physical concepts, methods, and techniques to human health. It spans four broad areas: medical imaging (creating pictures of internal structures), radiation therapy (using radiation to destroy tumours), biomechanics (applying mechanics to bones, joints, muscles, and circulation), and bioelectronics (understanding and manipulating the body's electrical signals).

Historically, physics drove medical revolutions. Wilhelm Röntgen's discovery of X-rays in 1895 gave physicians their first non-invasive window into the human body. The development of nuclear magnetic resonance in the 1940s led directly to MRI scanners

in the 1970s and 1980s. The physics of ultrasound, developed for submarine detection during World War II, became obstetric and cardiac ultrasound within two decades. Physics discoveries almost always find medical applications within a generation.

Historical Note: Röntgen discovered X-rays on 8 November 1895 while experimenting with cathode ray tubes. Within weeks he had produced the first X-ray image — of his wife's hand, showing her wedding ring and finger bones. Within a year, X-ray machines were being used in hospitals across Europe and North America. The time from physics discovery to medical deployment was less than twelve months.

54.2 Mechanics of the Human Body

The skeleton, muscles, and joints form a mechanical system. Bones act as levers, muscles provide forces, and joints act as fulcrums. Newton's laws apply: the body accelerates when net forces act on it and is in static equilibrium when the sum of forces and torques is zero.

Levers in the body are classified by the position of the fulcrum relative to the force and load. The elbow is a third-class lever: the bicep muscle (force) acts between the elbow joint (fulcrum) and the hand holding a load. Third-class levers sacrifice mechanical advantage for speed and range of motion — a small muscle contraction produces a large hand displacement.

$$\Sigma \mathbf{F} = 0 \quad \text{and} \quad \Sigma \tau = 0 \quad (\text{static equilibrium})$$

Conditions for static equilibrium applied to the body: net force = 0, net torque about any pivot = 0. These conditions allow calculation of joint reaction forces and muscle tensions.

Joint forces are often far larger than body weight because muscles attach close to joints (short moment arm). The hip joint, for example, carries forces of 2–3 times body weight during quiet standing and up to 8–9 times body weight during running, due to the short lever arms of the hip abductor muscles that counterbalance the body weight. This is why hip implant design requires understanding of biomechanical forces, not just material strength.

Example — Elbow lever analysis A 4.0 kg book is held in the palm, 35 cm from the elbow joint. The bicep attaches 5 cm from the joint. Find the bicep force. Torque balance: $F_{\text{bicep}} \times 0.05 = (4.0 \times 9.81) \times 0.35$. $F_{\text{bicep}} = 39.24 \times 0.35 / 0.05 = 274.7 \text{ N} \approx 275 \text{ N}$. The bicep exerts about seven times the weight of the book.

Bone is a composite material: a mineral phase (hydroxyapatite, providing compressive strength) surrounding an organic phase (collagen, providing tensile strength). Bone fails under different loads in different ways — it tolerates compression better than tension,

and tension better than shear. This is why ski fractures (shear and torsion), osteoporosis fractures (compressive failure of weakened bone), and stress fractures (repeated cyclic loading) are all mechanically distinct.

54.3 Biomechanics of Movement and Force

Biomechanics applies the principles of mechanics to biological systems, particularly the study of human movement, posture, gait, and sport. It combines Newtonian mechanics, material science, and anatomy to understand how forces are generated, transmitted, and absorbed by the body.

During walking, the centre of mass follows an inverted-pendulum path: the body vaults over the stance leg, converting kinetic energy to potential energy and back again. Running is a spring-mass system: the tendons (particularly the Achilles tendon and plantar fascia) store elastic potential energy during foot contact and release it during push-off, improving efficiency. The Achilles tendon stores and releases roughly 35% of the mechanical energy of each running stride.

Ground reaction forces (GRFs) are measured using force plates. During walking, the vertical GRF is approximately 1.2 times body weight at heel strike, drops to about 0.8 times body weight at midstance, and rises again to 1.2 times body weight at toe-off. During running, peak GRFs are 2–3 times body weight. These data inform rehabilitation, prosthetic design, and sports injury prevention.

Activity	Peak Vertical GRF (times body weight)	Notes
Quiet standing	1.0	Equal to body weight
Walking	1.0–1.3	Two peaks per step
Jogging	1.5–2.5	Increases with speed
Running (competitive)	2.0–3.0	Depends on surface and footwear
Jumping (landing)	3.0–8.0	Increases with drop height
Weightlifting (clean)	3.0–5.0	Bar acceleration phase

Table 54.1 *Typical peak vertical ground reaction forces for common activities.*

Injury occurs when forces exceed the structural limits of tissue. Stress (force per unit

area, $\sigma = F/A$) and strain (fractional deformation, $\epsilon = \Delta L/L$) describe the mechanical state of a tissue. Young's modulus $E = \sigma/\epsilon$ characterises stiffness. Bone has $E \approx 20$ GPa in compression; cartilage $\approx 1\text{--}5$ MPa; tendons $\approx 1\text{--}2$ GPa. Exceeding the ultimate tensile strength causes failure — fracture or rupture.

54.4 Pressure and Fluid Flow in Circulation

The cardiovascular system is a closed-loop hydraulic circuit. The heart is a dual pump: the right side pumps deoxygenated blood to the lungs (pulmonary circulation, low pressure); the left side pumps oxygenated blood to the body (systemic circulation, high pressure). Blood pressure is the gauge pressure exerted by blood on vessel walls.

Normal systemic blood pressure is reported as systolic / diastolic: typically 120 / 80 mmHg (16 / 10.7 kPa). Systolic pressure (120 mmHg) is the peak pressure during ventricular contraction. Diastolic pressure (80 mmHg) is the minimum pressure during ventricular relaxation. Blood pressure is measured with a sphygmomanometer (pressure cuff) and stethoscope, which detects the Korotkoff sounds caused by turbulent blood flow as the cuff pressure is released.

$$Q = \Delta P / R \quad (\text{Poiseuille flow in vessels})$$

Blood flow rate Q (m^3/s) = pressure difference ΔP (Pa) $\div$ vascular resistance R . Resistance $R = 8\mu L/(\pi r^4)$ where μ = blood viscosity, L = vessel length, r = radius. The r^4 dependence makes vessel radius the dominant factor in flow control.

The r^4 dependence of Poiseuille's law is medically critical. Halving the radius of a vessel (by atherosclerotic plaque, vasospasm, or stenosis) reduces flow by a factor of $2^4 = 16$ at the same pressure, or requires a 16-fold increase in pressure gradient to maintain the same flow. This is why arterial stenosis causes dramatically elevated blood pressure and why even partial blockages are clinically dangerous.

The continuity equation ($A_1 v_1 = A_2 v_2$) applies to blood flow: as vessels branch and total cross-sectional area increases, flow velocity decreases. Blood flows fastest in the aorta (≈ 20 cm/s peak) and slowest in the capillaries ($\approx 0.5\text{--}1.0$ mm/s), which allows adequate time for oxygen and nutrient exchange. The total cross-sectional area of all capillaries is about 500–1000 times that of the aorta.

Key Point: Blood pressure measured at the arm assumes the arm is at heart level. Raising the arm above heart level decreases the measured pressure (by ρgh); lowering it increases pressure. Clinicians always position the cuff at heart height for accurate readings.

54.5 Heat Transfer in Human Physiology

The human body is a thermodynamic system that generates heat through metabolism and must transfer it to the environment to maintain a stable core temperature of 37°C . The four mechanisms of heat transfer — conduction, convection, radiation, and evaporation — all operate simultaneously.

Conduction transfers heat through contact. The body's subcutaneous fat layer (poor conductor, $\kappa \approx 0.2 \text{ W/m}^{\circ}\text{C}$) provides thermal insulation. Clothing traps a layer of air (even poorer conductor, $\kappa \approx 0.025 \text{ W/m}^{\circ}\text{C}$), dramatically reducing conductive heat loss. In cold water, the body loses heat 25 times faster than in air at the same temperature because water is a much better conductor ($\kappa \approx 0.6 \text{ W/m}^{\circ}\text{C}$) and also forces convection over the skin surface.

Convection — heat loss through moving air or liquid — accounts for about 25–30% of heat loss at rest and increases with wind speed (wind chill) and exercise. Radiation accounts for about 40–60% of heat loss at rest — the body radiates as a near-perfect blackbody at infrared wavelengths. The power radiated is $P = \epsilon\sigma AT^4$, where $\epsilon \approx 0.97$ for human skin regardless of pigmentation. Evaporation of sweat is the dominant cooling mechanism during vigorous exercise, contributing up to 80% of heat loss.

Mechanism	% of Heat Loss at Rest	Key Factors	Increases With
Radiation	40–60%	Skin emissivity, skin-environment temp difference	High ambient temperature gradient
Convection	25–30%	Air velocity, skin-air temp difference	Wind speed, exercise
Conduction	2–3%	Contact area, material conductivity	Cold water immersion
Evaporation (sweat)	5–10% rest; up to 80% exercise	Sweat rate, humidity	Exercise intensity, humidity

Table 54.2 *Heat loss mechanisms in the resting human body at moderate ambient temperature.*

Fever is a controlled increase in the body's thermostat set point. Hyperthermia (uncontrolled overheating) is dangerous because enzyme activity is temperature-sensitive; at core temperatures above 41°C , protein denaturation begins. Hypothermia (core temperature below 35°C) slows metabolic rate dangerously. Clinically, heat is both used (therapeutic ultrasound, diathermy, laser ablation) and avoided (cooling blankets, ice packs) as part of treatment.

54.6 Electricity in the Nervous System and Heart

Neurons transmit information as electrical signals called action potentials. The resting membrane potential of a neuron is approximately -70 mV (inside negative relative to outside), maintained by the sodium-potassium pump and selective ion permeability. When stimulated, sodium ions rush in, depolarising the membrane; the action potential propagates along the axon as a self-regenerating wave at speeds of 1 – 100 m/s depending on axon diameter and myelination.

The heart has its own electrical pacemaker: the sinoatrial (SA) node generates spontaneous action potentials at about 60 – 80 times per minute, triggering a coordinated wave of contraction through the atria and ventricles. The ECG (electrocardiogram) records this electrical activity from electrodes on the skin. The characteristic P-QRS-T waveform maps to: P wave — atrial depolarisation; QRS complex — ventricular depolarisation (contraction); T wave — ventricular repolarisation.

Defibrillators deliver a brief high-energy electrical shock (200 – 360 J) to reset the heart's electrical activity during ventricular fibrillation (chaotic, uncoordinated contractions). The shock simultaneously depolarises all heart muscle cells, briefly stopping all electrical activity; the SA node (which has the fastest intrinsic rate) then resumes control and restores normal rhythm. Automated external defibrillators (AEDs) analyse rhythm and instruct even untrained users through the process.

Example — Defibrillator energy A defibrillator delivers 300 J of energy through paddles with contact resistance $50\ \Omega$ over 8 ms. The peak current: $E = I^2Rt \rightarrow I = \sqrt{E/Rt} = \sqrt{(300 / (50 \times 0.008))} = \sqrt{750} \approx 27.4$ A. This large transient current flows through the chest cavity, depolarising cardiac muscle.

Transcutaneous electrical nerve stimulation (TENS) uses low-frequency electrical currents to modulate pain signalling. Cochlear implants deliver precisely timed electrical pulses to auditory nerve fibres, bypassing damaged hair cells. Deep brain stimulation (DBS) uses implanted electrodes to treat Parkinson's disease, depression, and epilepsy. All of these applications depend on understanding the physics of bioelectricity.

54.7 Medical Imaging with X-Rays

X-rays are high-energy electromagnetic radiation (wavelength 0.01 – 10 nm, frequency 3×10^{13} – 3×10^{16} Hz). They are produced by bombarding a tungsten target with high-speed electrons accelerated through 30 – 150 kV. X-rays penetrate soft tissue but are absorbed by denser materials (bone, metal). The transmitted X-rays darken

photographic film or activate a digital detector, producing a shadow image.

X-ray absorption follows Beer's law: $I = I_0 e^{(-\mu x)}$, where I_0 is incident intensity, x is the thickness of material, and μ is the linear attenuation coefficient. Different tissues have different μ values: bone (high μ , absorbs strongly, appears white on film), lung tissue (low μ , appears dark), and soft tissue (intermediate μ , grey). Contrast agents — iodine compounds injected into blood vessels, or barium swallowed to outline the gastrointestinal tract — increase contrast by raising μ in specific regions.

$$I = I_0 e^{(-\mu x)}$$

X-ray attenuation (Beer's law). I = transmitted intensity; I_0 = incident intensity; μ = linear attenuation coefficient (m^{-1}); x = thickness of material (m). Each material has a characteristic μ that depends on density and atomic number.

Computed Tomography (CT) uses a rotating X-ray source and detector array to acquire hundreds of shadow projections around the patient. Computer algorithms (filtered back-projection or iterative reconstruction) combine these projections to compute a three-dimensional map of X-ray attenuation coefficients throughout the body, which can be visualised as cross-sectional slices or three-dimensional volume renders. CT greatly improves the detection of soft-tissue structures compared to plain X-ray.

The biological risk of X-rays arises from ionisation: X-ray photons have enough energy to eject electrons from atoms, breaking chemical bonds and potentially damaging DNA. The radiation dose is measured in gray (Gy, 1 Gy = 1 J/kg absorbed) and effective dose in sievert (Sv), which weights absorbed dose by tissue sensitivity. A chest X-ray delivers about 0.1 mSv effective dose; a CT scan of the abdomen delivers about 10 mSv — comparable to the annual natural background dose.

Common Mistake: X-ray and CT scans deliver ionising radiation. The risk is small for individual scans but accumulates with repeated exposure. Clinicians use the ALARA principle (As Low As Reasonably Achievable) to minimise dose while obtaining diagnostically adequate images.

54.8 MRI and Magnetic Field Applications

Magnetic Resonance Imaging (MRI) exploits nuclear magnetic resonance (NMR): the tendency of atomic nuclei with non-zero spin to align with an external magnetic field and resonate at a characteristic frequency when disturbed. In clinical MRI, hydrogen nuclei (protons) in water and fat are the signal source.

A patient is placed inside a superconducting magnet generating a uniform field of 1.5–3.0 T (about 30 000–60 000 times Earth's field). Protons align with this field. A brief radiofrequency (RF) pulse at the Larmor frequency ($\omega = \gamma B$, where γ is the

gyromagnetic ratio for hydrogen = 42.58 MHz/T) tips the protons out of alignment. As they relax back, they emit RF signals detected by receiver coils. The decay time constants T_1 and T_2 differ between tissues (fat, water, grey matter, white matter, tumour), providing the contrast that makes MRI so powerful for soft-tissue imaging.

$$\omega_0 = \gamma B_0$$

Larmor equation. ω_0 = resonance (Larmor) frequency (rad/s); γ = gyromagnetic ratio (42.58 MHz/T for ^1H); B_0 = static magnetic field strength (T). The RF pulse must exactly match ω_0 to excite proton resonance.

Gradient coils superimpose small linear variations on the main field, making B_0 (and therefore ω_0) position-dependent. By varying these gradients in three directions and using Fourier analysis of the detected signals, a complete three-dimensional spatial map of proton density and relaxation times is reconstructed. No ionising radiation is involved: MRI uses RF radiation, which is non-ionising and considered biologically safe at clinical field strengths.

MRI is contraindicated in patients with ferromagnetic implants (certain pacemakers, cochlear implants, surgical clips, metal fragments) because the strong magnetic field exerts forces and torques on ferromagnetic objects, and the RF pulse can induce dangerous currents in conducting implants. The MRI suite must be a controlled zone with strict metallic object exclusion.

54.9 Ultrasound and Sound Wave Imaging

Medical ultrasound uses high-frequency sound waves (1–20 MHz, well above the 20 kHz upper limit of human hearing) to create images of internal structures. A piezoelectric transducer converts electrical pulses to ultrasound pulses and, in between pulses, detects reflected sound (echoes) returning from tissue boundaries.

Echoes are produced wherever sound encounters a change in acoustic impedance $Z = \rho v$, where ρ is tissue density and v is the local speed of sound. The reflection coefficient $R = (Z_2 - Z_1)^2 / (Z_2 + Z_1)^2$ determines what fraction of the incident intensity is reflected. Strong reflections occur at tissue-air interfaces (lung, bowel gas) and tissue-bone interfaces; weaker reflections occur at boundaries between similar soft tissues. The time delay between transmitted pulse and received echo gives depth: $\text{depth} = (v \times t) / 2$.

$$d = vt / 2$$

Ultrasound depth calculation. d = depth of reflecting surface (m); v = speed of sound in tissue (≈ 1540 m/s in soft tissue); t = round-trip travel time (s). The factor of 2 accounts for the pulse travelling to and from the reflector.

B-mode (brightness mode) ultrasound displays a two-dimensional cross-sectional image by sweeping the beam across a plane and plotting echo strength as pixel brightness. Doppler ultrasound uses the Doppler shift of reflected sound to measure blood flow velocity and direction — blood moving toward the transducer shifts the frequency up; blood moving away shifts it down. Colour Doppler maps flow direction and relative speed across a two-dimensional image in real time.

Ultrasound is safe (no ionising radiation), portable, inexpensive relative to MRI/CT, and provides real-time imaging. It is the first-choice modality for obstetric imaging, cardiac imaging (echocardiography), guiding needle biopsies and injections, and imaging soft-tissue structures such as the liver, kidneys, thyroid, and tendons. Its main limitations are poor penetration through gas and bone, and operator dependence.

Example — Ultrasound depth An echo is received 130 μs after a pulse is transmitted. Speed of sound in tissue = 1540 m/s. Depth = $1540 \times 130 \times 10^{-6} / 2 = 0.1001 \text{ m} \approx 10 \text{ cm}$. The reflecting surface is about 10 cm below the transducer.

54.10 Radiation Therapy and Nuclear Medicine

Radiation therapy uses ionising radiation to destroy cancer cells. High-energy X-rays (4–25 MV, produced by a linear accelerator), gamma rays (from ^{137}Cs or ^{60}Co sources), protons, or heavy ions are directed at tumours. The radiation damages DNA, preventing cancer cells from dividing. Because cancer cells divide rapidly and have impaired DNA repair mechanisms, they are more radiosensitive than most normal tissues.

Modern radiation therapy uses intensity-modulated radiation therapy (IMRT) and image-guided radiation therapy (IGRT) to deliver shaped beams from multiple angles that converge on the tumour while sparing surrounding tissue. Stereotactic radiosurgery (Gamma Knife, CyberKnife) delivers hundreds of low-intensity beams that intersect only at the tumour, achieving surgical precision without a physical incision.

Nuclear medicine uses radioactive isotopes administered to the patient (as tracers or therapeutic agents). Diagnostic nuclear medicine: a gamma-emitting isotope (typically $^{99\text{m}}\text{Tc}$, technetium-99m, with a 6-hour half-life) is attached to a biologically active molecule and injected. The molecule preferentially accumulates in specific organs or tumours. A gamma camera detects the emitted photons and produces a functional image showing where the tracer has concentrated.

Positron Emission Tomography (PET) uses positron-emitting isotopes (^{18}F is most common, with a 110-minute half-life). Emitted positrons immediately annihilate with electrons, producing two back-to-back 511 keV gamma photons. Coincidence detection

of both photons identifies the line of emission; reconstructing many such lines produces a three-dimensional image of tracer concentration. FDG-PET (fluorodeoxyglucose) images glucose metabolism — cancer cells, which metabolise glucose at high rates, appear as bright spots.

Modality	Type of Radiation	Purpose	Key Isotope / Source
Planar scintigraphy	Gamma rays	Organ function imaging	^{99m}Tc
SPECT	Gamma rays (tomographic)	3D functional imaging	^{99m}Tc , ^{123}I
PET	Annihilation photons (511 keV)	Metabolic imaging	^{18}F -FDG, ^{13}N
Radioiodine therapy	Beta particles	Thyroid cancer treatment	^{131}I
IMRT / IGRT	High-energy X-rays	Tumour destruction	Linear accelerator
Brachytherapy	Gamma/beta (short range)	Internal tumour treatment	^{125}I , ^{192}Ir

Table 54.3 Common radiation therapy and nuclear medicine modalities.

54.11 Prosthetics and Biomedical Engineering

Biomedical engineering applies engineering principles to medicine and biology. Prosthetics — artificial replacement limbs — require careful biomechanical analysis. A lower-limb prosthetic must replicate the mechanical function of the missing joint and limb segment, including load bearing, energy storage and return during gait, and proprioceptive feedback.

Modern running prosthetics (the Cheetah® blade used by para-athletes) are made from carbon-fibre composites designed to store elastic strain energy during stance and release it during push-off, mimicking the Achilles tendon. The spring stiffness is tuned to the athlete's mass and running speed. Osseointegration — direct anchoring of the prosthetic to a titanium implant that fuses with the bone — improves load transfer compared to socket-based attachment.

Cochlear implants bypass damaged hair cells in the inner ear. An external microphone captures sound; a signal processor decomposes it into frequency channels; an implanted electrode array delivers current pulses to different locations along the spiral cochlea (which encodes frequency tonotopically — high frequencies at the base, low frequencies at the apex). The timing and pattern of pulses encode speech and music information directly to the auditory nerve.

Biomedical sensors are ubiquitous in modern healthcare. Pulse oximeters use the different optical absorption spectra of oxygenated and deoxygenated haemoglobin at two wavelengths (660 nm red and 940 nm infrared) to measure blood oxygen saturation. Continuous glucose monitors use electrochemical sensors implanted subcutaneously. Implantable cardiac monitors record ECG data continuously for months to detect arrhythmias.

54.12 Safety and Ethics in Medical Technology

Physical technologies in medicine carry risks that must be rigorously managed. Ionising radiation causes DNA damage; cumulative doses are tracked and limited for patients, radiation workers, and the general public. International Commission on Radiological Protection (ICRP) guidelines set dose limits: 20 mSv/year for occupationally exposed workers and 1 mSv/year for members of the public from artificial sources.

MRI strong magnetic fields can accelerate ferromagnetic objects into the magnet bore (projectile risk), induce currents in conducting implants, and heat metal near the field. Ultrasound can cause bioeffects (cavitation and local heating) at high intensities; diagnostic ultrasound operates well below these thresholds. Lasers can damage the eyes and skin. Electrical devices must be electrically isolated to prevent microshock (leakage current as low as 10 μ A can induce ventricular fibrillation if a catheter carries it directly to the heart).

Ethical considerations include equitable access to imaging technology (MRI scanners are concentrated in wealthy countries and urban centres), the use of AI-based image interpretation (accuracy, bias, accountability), genetic data from nuclear medicine and imaging biobanks, and the environmental disposal of radioactive waste from nuclear medicine. Physics students who become healthcare workers, engineers, or policymakers will face these questions directly.

Study Tip: When answering questions about medical imaging, always identify: (1) what type of radiation or signal is used, (2) what physical property of tissue creates contrast, and (3) what the biological safety concerns are. These three dimensions distinguish X-ray, CT, MRI, ultrasound, PET, and nuclear scintigraphy from each other.

54.13 Common Mistakes in Medical Physics Interpretation

- Confusing ionising and non-ionising radiation. X-rays and gamma rays are ionising (enough energy per photon to eject electrons). UV light, visible light, infrared, microwaves, and RF (including MRI pulses) are non-ionising. Many students incorrectly assume all medical radiation is ionising and dangerous. MRI and ultrasound do not use ionising radiation.
- Assuming higher magnetic field is always better in MRI. Higher field (3 T vs 1.5 T) improves signal-to-noise ratio but also increases artefacts, cost, contraindications, and specific absorption rate (RF heating). 1.5 T remains the most widely used clinical field strength for most applications.
- Forgetting the round-trip factor in ultrasound depth calculations. $\text{Depth} = v \times t / 2$. The echo must travel to the reflector and back, so dividing by 2 is required. Omitting it gives double the actual depth.
- Thinking Doppler ultrasound measures absolute velocity. It measures the component of velocity along the beam direction. If the beam is perpendicular to flow, the Doppler shift is zero regardless of speed. Clinicians correct for the angle between the beam and the flow direction.
- Confusing gray (Gy) and sievert (Sv). Gray measures absorbed dose (energy per mass); sievert measures effective dose (weighted by radiation type and tissue sensitivity). For X-rays and gamma rays, $1 \text{ Gy} = 1 \text{ Sv}$. For alpha particles (more biologically damaging), $1 \text{ Gy} = 20 \text{ Sv}$. Do not interchange units.
- Applying Poiseuille's law uncritically to blood flow. Blood is non-Newtonian (its viscosity changes with shear rate), the vessels are elastic (not rigid), and flow is pulsatile (not steady). Poiseuille's law is a useful approximation but must not be applied as an exact description of real circulatory physics.

Common Mistake: Radiation dose units are frequently confused in medical contexts. Gray (Gy) is absorbed dose; sievert (Sv) is effective dose. For gamma rays and X-rays the numerical values are equal, but for alpha radiation $1 \text{ Gy} = 20 \text{ Sv}$. Using the wrong unit can misrepresent risk by a factor of 20.

Chapter Summary

Medical physics applies mechanics, fluid dynamics, thermodynamics, electricity, waves, and nuclear physics to the human body and to medical technology. Physical principles underlie every major diagnostic and therapeutic tool in modern medicine.

- Biomechanics: the body is a lever system in static equilibrium; muscles exert forces far greater than body weight due to short moment arms; bone, tendon, and cartilage obey stress-strain relationships.
- Circulation: blood pressure, Poiseuille flow, continuity, and Bernoulli's principle govern cardiovascular fluid dynamics; arterial stenosis dramatically increases resistance (r^4 dependence).
- Heat: the body loses heat by radiation (40–60%), convection (25–30%), conduction, and evaporation; temperature regulation is critical for enzyme function.
- Bioelectricity: neurons and the heart generate and transmit electrical signals; ECG, EEG, pacemakers, defibrillators, and TENS all exploit bioelectrical principles.
- X-rays and CT: ionising EM radiation; contrast from differential tissue attenuation; dose measured in Gy and Sv; CT reconstructs 3D images from multiple projections.
- MRI: non-ionising RF and strong magnetic field; contrast from proton density and T1/T2 relaxation; no radiation risk; strong contraindications for ferromagnetic implants.
- Ultrasound: non-ionising sound (1–20 MHz); depth from echo timing ($d = vt/2$); Doppler measures flow; safe for obstetric and cardiac imaging.
- Nuclear medicine and radiation therapy: radioactive tracers for functional imaging; high-energy beams for tumour destruction; dose carefully controlled.
- Prosthetics and biomedical engineering: carbon-fibre prosthetics store and return elastic energy; cochlear implants deliver electrical pulses to auditory nerve; sensors measure O_2 saturation, glucose, and ECG continuously.

Key Terms

biomechanics: The application of mechanical principles to biological systems, particularly to the forces, stresses, and motions involved in human movement and the function of the musculoskeletal system.

lever (biological): A rigid structure (bone) that rotates about a fulcrum (joint) under the action of a muscle force to move a load; classified as first, second, or third class by the relative position of force, fulcrum, and load.

ground reaction force (GRF): The force exerted by the ground on the body in reaction to the body's weight and acceleration; equal and opposite to the force exerted by the body on the ground (Newton's third law).

blood pressure: The gauge pressure of blood on artery walls; reported as systolic/diastolic in mmHg (normal: 120/80 mmHg); measured with a sphygmomanometer.

Poiseuille's law: $Q = \pi r^4 \Delta P / (8\mu L)$; the volume flow rate through a cylindrical vessel is proportional to the fourth power of the radius, making vessel radius the dominant factor in flow resistance.

action potential: A brief, self-propagating electrical signal in a neuron or muscle cell; produced by rapid depolarisation (Na^+ influx) and repolarisation (K^+ efflux) of the cell membrane.

ECG (electrocardiogram): A recording of the electrical activity of the heart from surface electrodes; the P-QRS-T waveform corresponds to atrial depolarisation, ventricular depolarisation, and ventricular repolarisation.

X-ray attenuation (Beer's law): $I = I_0 e^{(-\mu x)}$; the exponential reduction of X-ray intensity as it passes through material; μ depends on tissue density and atomic number.

Larmor frequency: $\omega_0 = \gamma B_0$; the resonance frequency of proton spins in an external magnetic field; the basis of MRI signal excitation.

acoustic impedance: $Z = \rho v$; the product of tissue density and sound speed; determines the fraction of ultrasound reflected at tissue boundaries.

absorbed dose (Gy): Energy deposited by radiation per unit mass of tissue; $1 \text{ Gy} = 1 \text{ J/kg}$; does not account for biological effectiveness of different radiation types.

effective dose (Sv): Absorbed dose weighted by radiation type and tissue sensitivity; accounts for biological risk; $1 \text{ Sv} = \text{biologically equivalent risk to } 1 \text{ Gy of X-rays}$.

PET (Positron Emission Tomography): Nuclear medicine imaging using positron-emitting isotopes; annihilation produces two 511 keV photons detected in coincidence to localise tracer concentration.

ALARA: As Low As Reasonably Achievable; the guiding principle of radiation protection: minimise dose while maintaining diagnostic or therapeutic benefit.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain why the bicep muscle must exert a force much greater than the weight of an object held in the hand. What mechanical property of the arm produces this force amplification?

2. A patient has significant narrowing (stenosis) of a coronary artery. Explain, using Poiseuille's law, why even a 50% reduction in radius (by area, approximately) would severely compromise blood flow.
3. Distinguish between the information provided by an X-ray and an MRI scan of the same body region. What physical property of tissue creates contrast in each modality?
4. Why is ultrasound preferred over X-ray for imaging a developing fetus? What limitation does ultrasound have compared to X-ray?
5. Explain how a PET scan can detect a tumour that is too small to be seen on a CT scan. What physical process underlies the PET signal?
6. A patient with a steel orthopaedic implant is referred for MRI. The radiologist declines. Why? What physical hazard is the concern?
7. Why does blood pressure measured at the ankle differ from blood pressure measured at the arm in a person standing upright? Quantify the difference using the hydrostatic pressure formula if the vertical distance is 1.2 m.
8. Explain the ALARA principle. Give one specific example of a clinical decision guided by ALARA.

Biomechanics Problems

9. A person holds a 5.0 kg weight in the hand. The elbow is flexed at 90° . The bicep attaches 4.5 cm from the elbow joint; the weight is 33 cm from the joint. Find the bicep force. Find the joint reaction force (ignore forearm weight).
10. A 70 kg person stands on one leg during walking. The hip abductor muscles must balance the body weight about the hip joint. If the abductor force arm is 6.0 cm and the body's centre of mass is 13 cm from the hip joint (on the opposite side), find the abductor muscle force.
11. A tibia (shin bone) with cross-sectional area 3.2 cm^2 supports a compressive force of 2500 N. Find the compressive stress in MPa. If the bone fractures at 150 MPa, what is the safety factor?
12. During running, the Achilles tendon experiences a force of 2800 N. The tendon has a cross-sectional area of 80 mm^2 . Find the tensile stress. If the tendon's ultimate tensile strength is 70 MPa, what fraction of the failure load is being applied?

13. A bone is modelled as a circular cylinder of length 0.30 m and cross-sectional area 4.0 cm^2 . A compressive load of 1500 N shortens it by 0.06 mm. Find Young's modulus for the bone.

Medical Imaging Analysis

14. An X-ray beam with initial intensity I_0 passes through 8.0 cm of soft tissue ($\mu = 0.22 \text{ cm}^{-1}$) followed by 1.5 cm of bone ($\mu = 2.4 \text{ cm}^{-1}$). Find the transmitted intensity as a fraction of I_0 .
15. An MRI scanner operates at 3.0 T. Find the Larmor frequency for hydrogen protons in MHz. ($\gamma = 42.58 \text{ MHz/T}$)
16. An ultrasound echo returns from a kidney 142 μs after the pulse is transmitted. Find the depth of the kidney surface. ($v = 1540 \text{ m/s}$ in soft tissue)
17. A Doppler ultrasound probe measures a frequency shift of 5.2 kHz from blood flow. The probe operates at 7.5 MHz and makes an angle of 60° with the vessel. Find the blood flow velocity. (Use $\Delta f/f = 2v \cdot \cos\theta/c$, $c = 1540 \text{ m/s}$)
18. A patient receives a chest CT scan delivering 8.0 mGy of X-radiation to the lungs. If the radiation weighting factor for X-rays is 1 and the tissue weighting factor for lungs is 0.12, find the effective dose in mSv.

Radiation and Dosage Exercises

19. A radiation worker receives a whole-body effective dose of 3.2 mSv from X-rays and 0.8 mSv from neutrons (radiation weighting factor = 20, mean absorbed dose from neutrons = 0.04 mGy). The annual limit is 20 mSv. What fraction of the annual limit has the worker received?
20. A $^{99\text{m}}\text{Tc}$ source starts with activity 740 MBq. Its half-life is 6.0 hours. Find the activity after 24 hours. Why does the short half-life make it suitable for diagnostic imaging?
21. A radiation therapy treatment delivers 50 Gy to a tumour over 25 fractions. Find the dose per fraction. If the tumour is in the lung (tissue weighting factor 0.12) and X-rays are used (weighting factor 1), find the effective dose per fraction.

22. Compare the effective dose from: (a) a dental X-ray (0.005 mSv), (b) a chest X-ray (0.1 mSv), (c) an abdomen CT (10 mSv), and (d) annual natural background dose (2.4 mSv). Arrange in order and comment on the relative risks.

Applied Biomedical Physics Problems

23. A sphygmomanometer cuff is placed on the upper arm at heart level. The systolic pressure reads 125 mmHg. If the cuff were placed on the arm 20 cm above heart level, what would the reading be? (1 mmHg = 133 Pa; $\rho_{\text{blood}} = 1060 \text{ kg/m}^3$)
24. A patient's heart outputs 5.5 L/min at a mean arterial pressure of 95 mmHg. Compute the total peripheral vascular resistance in Pa·s/m³. (Convert all units to SI first.)
25. A defibrillator stores 350 J of energy in a capacitor and discharges through a transthoracic resistance of 70Ω in 10 ms. (a) Find the peak current assuming constant discharge. (b) Find the peak voltage. (c) Is this a fatal current for a person whose ventricular fibrillation threshold is 100 mA? Explain.
26. The cochlea responds to frequencies from 20 Hz to 20 000 Hz. A cochlear implant electrode array has 22 electrodes distributed along 25 mm of the cochlea. Assuming linear frequency mapping along the cochlea (logarithmic in frequency), find the frequency assigned to the electrode at 12.5 mm from the base. (Base: 20 000 Hz; apex: 20 Hz; use logarithmic interpolation.)

Chapter 54 Checkpoint

Before moving on to Chapter 55, confirm you can do each of the following.

I can...	See Section
Explain how physics principles underlie major medical diagnostic and therapeutic tools	54.1
Analyse the body as a lever system and find muscle forces using torque balance	54.2–54.3

Apply Poiseuille's law to blood flow and explain the effect of vessel radius	54.4
Identify the four mechanisms of heat loss from the body and rank their contributions	54.5
Describe action potentials, ECG waveforms, and the operation of defibrillators	54.6
Explain X-ray image formation and apply the attenuation law $I = I_0 e^{-\mu x}$	54.7
Describe MRI signal generation using the Larmor equation and T1/T2 relaxation	54.8
Apply $d = vt/2$ to calculate ultrasound depth and explain Doppler flow measurement	54.9
Distinguish between nuclear medicine imaging and radiation therapy	54.10
Explain the physics of prosthetics, cochlear implants, and pulse oximetry	54.11
Convert between absorbed dose (Gy) and effective dose (Sv) correctly	54.12
Identify and avoid common mistakes in medical physics problems	54.13

Continue to Chapter 55: Physics of Earth and Atmosphere →